CHE210: *DISCUSSION SESSION 13* **NAME:**___________________________________

NOVEMBER 26, 2019

Liquid sulfur at 130°C and pure oxygen at 25°C are mixed at a stoichiometric ratio and fed to a burner, where oxidation of sulfur to SO_2 takes place. The burner is perfectly insulated, the pressure is 1 atm, and the system operates at steady state. What is the temperature of the exiting stream?

<u>*Useful information:*</u>

$$S_{(s)} + O_{2\,(g)} \rightarrow SO_{2\,(g)}\ ;\ \text{where } \Delta H_r^{\circ} = -296.8\frac{kJ}{gmol}$$

$$\Delta\widehat{H}_m = 1.73\frac{kJ}{gmol}\ \text{for S at } 114℃$$

$$C_{p,S(l)} = 32\frac{J}{gmol\cdot ℃} \qquad C_{p,SO_2(g)} = 39.9\frac{J}{gmol\cdot ℃}$$

$$C_{p,S(s)} = 23.2\frac{J}{gmol\cdot ℃} \qquad C_{p,O_{2(g)}} = 29.3\frac{J}{gmol\cdot ℃}$$